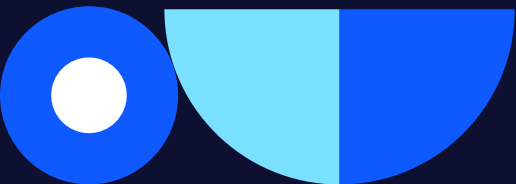




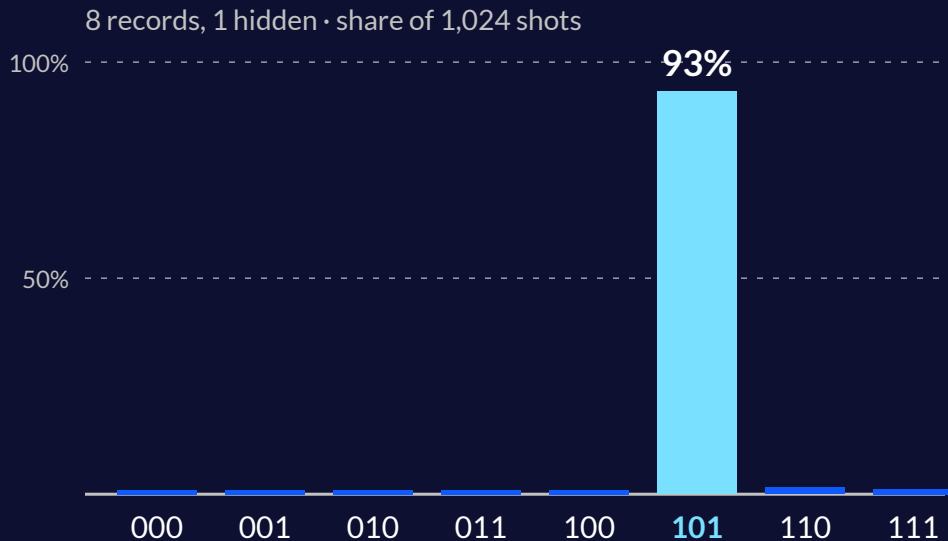
Quantum × Big Data

Where it helps, where it breaks, and where the hardware stands in 2026

Part 3 of 3 · Phúc · Big Data seminar



From the demo to real data

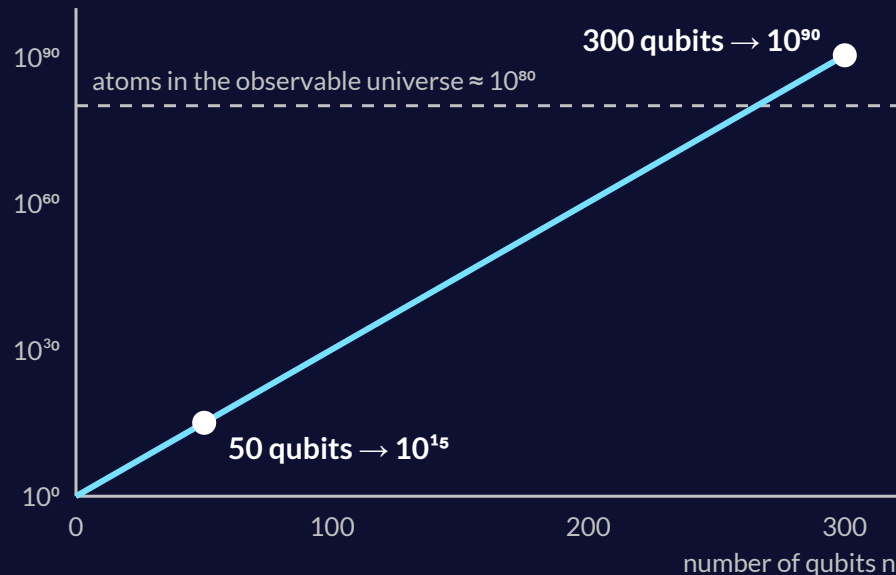


**Grover found it
without checking the
records one by one.**

Now scale that to a table with 10^{12} rows. What does quantum computing actually buy a big-data system?

Our Grover demo: 3 qubits, 2 iterations, 93% of 1,024 shots land on the hidden record 101.

Why quantum meets big data



2^{50}

$\approx 1.1 \times 10^{15}$ amplitudes, described by just 50 qubits

50 qubits $\rightarrow 10^{15}$ amplitudes; 300 qubits $\rightarrow 10^{90}$, above the $\approx 10^{80}$ atoms line.

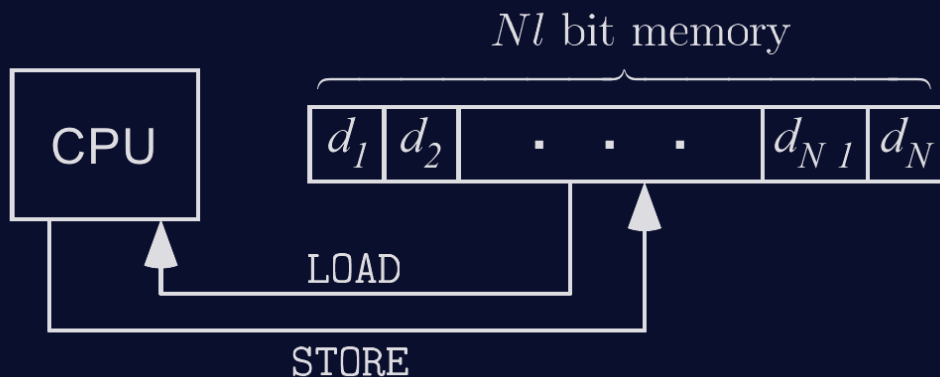


Data task → quantum tool

Big data task	Quantum building block
Record lookup	Grover search, $O(\sqrt{N})$
Classification and learning	Quantum kernels, variational circuits
Regression, PCA, linear systems	HHL algorithm
Scheduling, feature selection, clustering	QAOA, quantum annealing



Quantum search over a database



Unsorted N records: classical search needs $O(N)$ LOADs,
Grover needs $O(\sqrt{N})$.

Classical model: the CPU reaches memory only through LOAD and STORE. Source:
Nielsen & Chuang (2010), Fig. 6.8.

For $N = 10^{12}$ records

5×10^{11}

classical lookups on average (unsorted)

7.9×10^5

Grover iterations, $(\pi/4) \cdot \sqrt{N}$

≈ 40

comparisons with a sorted index ($\log_2 N$), faster than Grover



Quantum machine learning

Approach 1

Quantum kernels (QSVM)

- Encode data x as a quantum state $|\varphi(x)\rangle$
- The quantum chip estimates similarity $K(x, x') = |\langle \varphi(x) | \varphi(x') \rangle|^2$
- A classical SVM does the rest (Havlíček et al., Nature 2019)

We ran one: results on the next slide

Approach 2

Variational circuits (VQC)

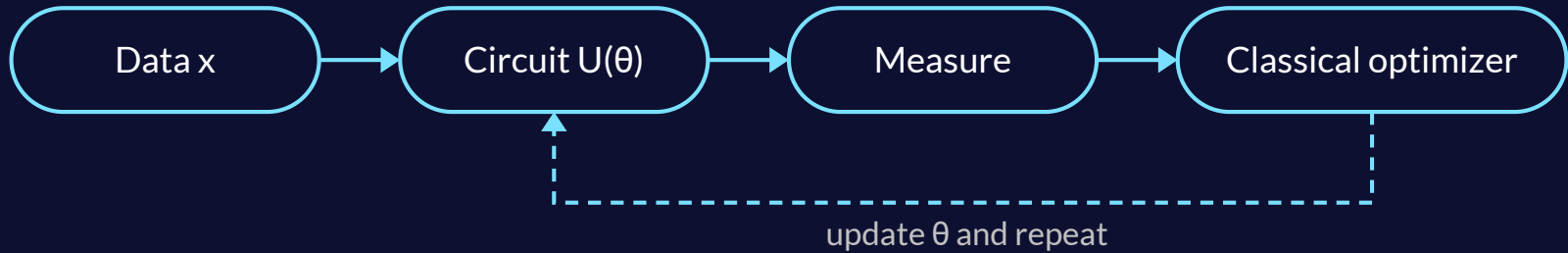
runs today

- Gates have trainable angles θ , like network weights
- The quantum chip runs the circuit; a classical optimizer updates θ
- Shallow circuits make it practical on NISQ devices

Try it: Qiskit Machine Learning, PennyLane

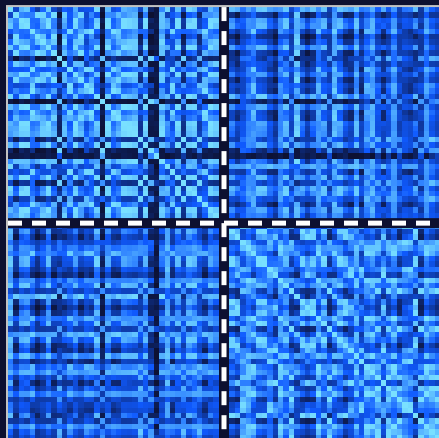


The hybrid loop



Data $x \rightarrow$ Circuit $U(\theta) \rightarrow$ Measure $\rightarrow$ Classical optimizer $\rightarrow$ Update θ , repeat

A real quantum kernel, measured

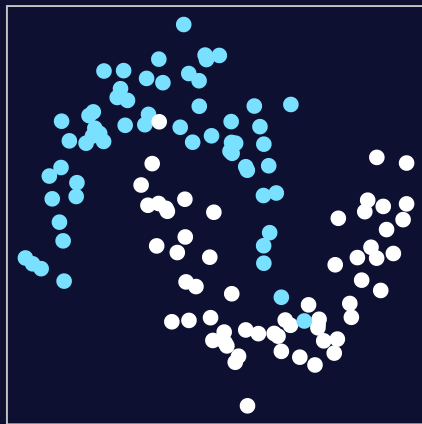


0

0.5

1

Training kernel, sorted by class; the colour scale shows similarity. Dashed lines split the two classes.



● class 0

● class 1

120 points, two interleaved moons · 80 train / 40 test · 2-qubit ZZ feature map on a simulator, same tuning for every kernel

Test accuracy · 40 held-out points

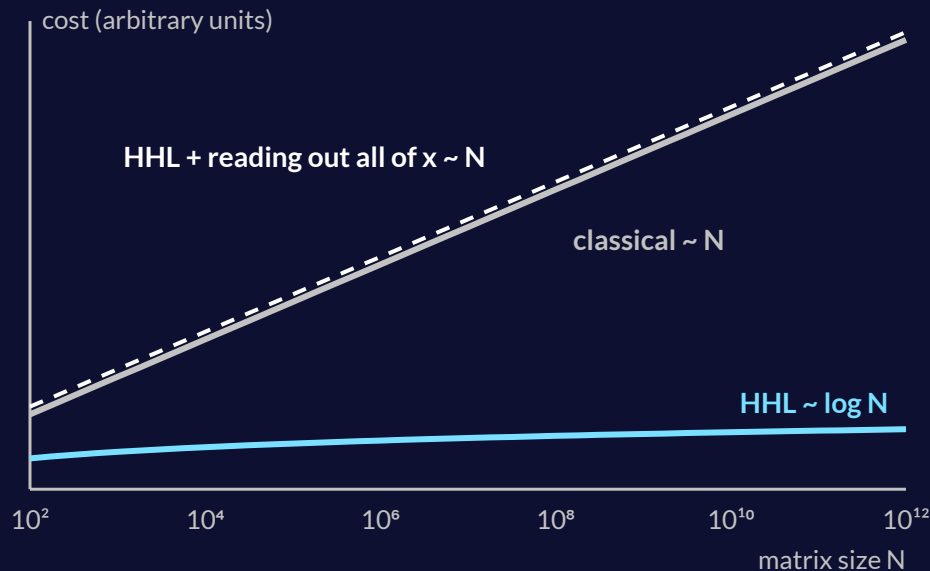
Quantum kernel (ZZ) **90.0%**

Linear SVM **92.5%**

RBF SVM **97.5%**

On classical data, the quantum kernel did not beat a tuned classical one.
“Compared to what?” matters.

HHL: exponential speedup, on paper



s = sparsity $\cdot$ κ = condition number $\cdot$ ϵ = precision

Classical $\cdot$ conjugate gradient

$$O(N \cdot s \cdot \kappa)$$

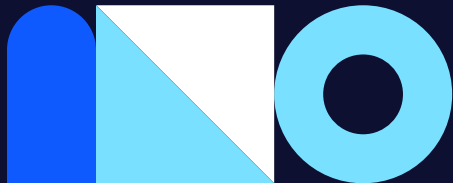
Quantum $\cdot$ HHL, 2009

$$O(\log N \cdot s^2 \kappa^2 / \epsilon)$$

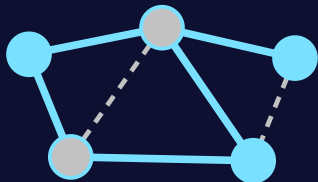


Four conditions attached

- A must be sparse and well-conditioned (small κ)
- The vector b must load into a quantum state $|b\rangle$ quickly
- The output is a state $|x\rangle$, not the vector: reading all N entries costs $O(N)$
- Useful only when you need a summary value such as $\langle x|M|x\rangle$



Quantum optimization



The problem

Many data tasks are QUBO problems

Feature selection, clustering, scheduling and routing all explode combinatorially.



Gate-based

QAOA (Farhi et al., 2014)

A hybrid, shallow-circuit algorithm designed for today's noisy hardware.



Special-purpose

Quantum annealing (D-Wave)

Special-purpose hardware with thousands of qubits, but not a general-purpose computer.

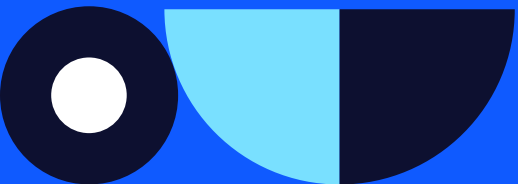
Reality check: no clear advantage yet over the best classical heuristics on real-world problems.



The catch

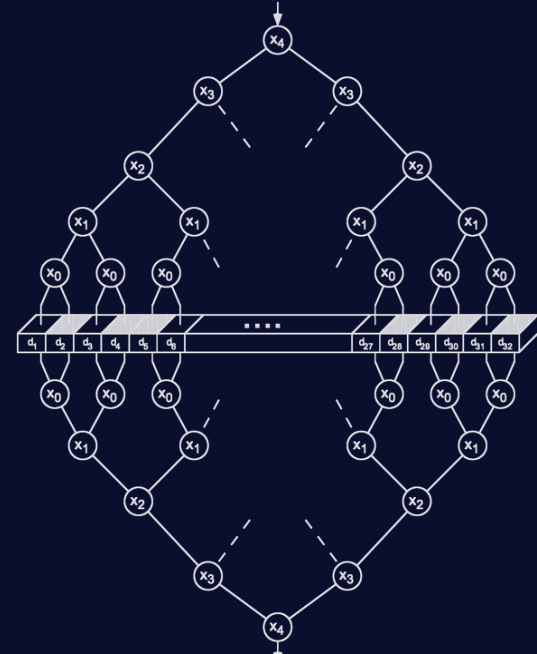
**Every speedup so far assumes
the data is already inside the
quantum computer.**

So how does big data get in?

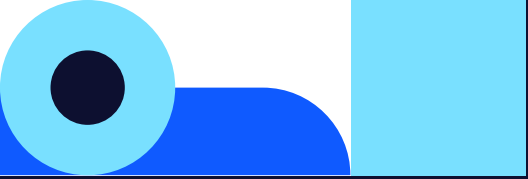


Bottleneck 1: loading the data

- Big data lives on disks as classical bits, while the algorithms assume it is already a quantum state.
- That needs **QRAM**: memory addressed by an index held in superposition.
- The textbook design uses $O(N \log N)$ quantum switches, roughly the hardware of the database itself.
- Loading N records alone costs about $O(N)$, erasing a $\sqrt{N}$ or $\log N$ speedup. No large, noise-tolerant QRAM exists yet.

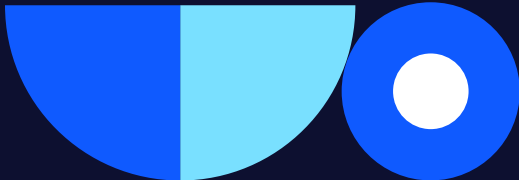


Quantum addressing for a 32-cell memory. Source: Nielsen & Chuang (2010), Fig. 6.9.



Nielsen & Chuang conclude that
Grover's main use is likely solving hard
search problems such as SAT or TSP, not
searching classical databases.

—Nielsen & Chuang (2010), §6.5





Bottleneck 2: the fine print

2009

HHL

Exponential speedup for linear systems, under strict conditions.

2015

Scott Aaronson

“Read the fine print” in Nature Physics lists the input, matrix and output caveats.

2016

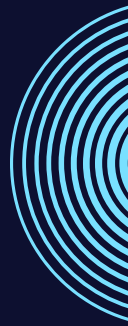
Kerenidis–Prakash

A quantum recommendation system, hailed as an exponential QML speedup.

2018

Ewin Tang

An 18-year-old student finds a classical algorithm nearly as fast, given the same data access.



Lesson: give the classical side the same assumptions about the data, and the quantum advantage often shrinks. Many QML algorithms were later “dequantized” the same way.





Hardware in 2026

From NISQ (noisy, intermediate-scale) toward early fault tolerance: many noisy physical qubits plus error correction make one reliable logical qubit.

Company	Technology	Milestone
Google	Superconducting	Willow, 105 qubits (Dec 2024): errors drop as the code grows, below threshold
IBM	Superconducting	Nighthawk, 120 qubits; Starling target (2029): 200 logical qubits, 100M gates
IonQ, Quantinuum	Trapped ion	Highest two-qubit gate fidelities, reported up to 99.99%
Pasqal, QuEra	Neutral atom	Hundreds of qubits; May 2026: logical beats physical qubits on a differential-equation task



Policy: NIST post-quantum cryptography standards (2024) · a US order targets a fault-tolerant machine by 2028 (June 2026)



Hardware in 2026

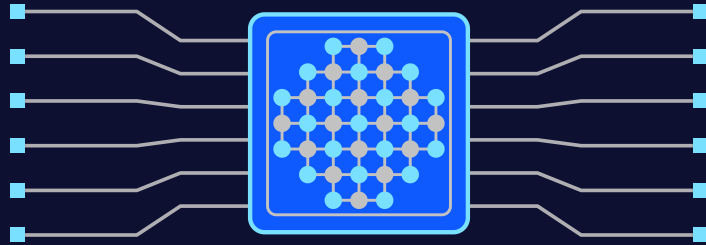
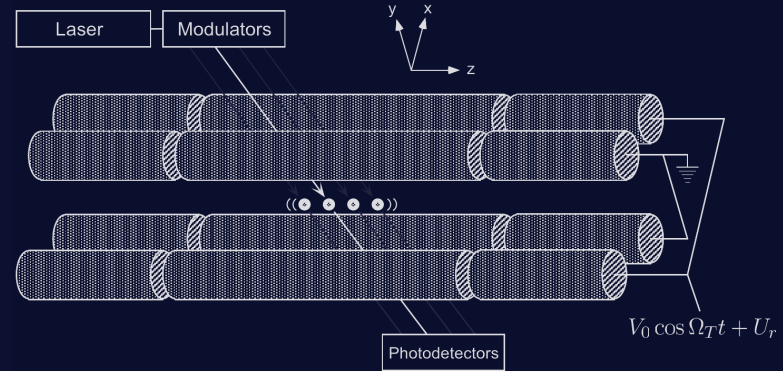


Illustration: superconducting chip



Trapped ions · Nielsen & Chuang, Fig. 7.7

Three takeaways

A co-processor, not a replacement

Quantum computers win only on narrow problem classes: simulation, factoring, some search and optimization.

The bottleneck is data I/O

For big data, the hard part is getting data in and out of the machine, not computing fast.

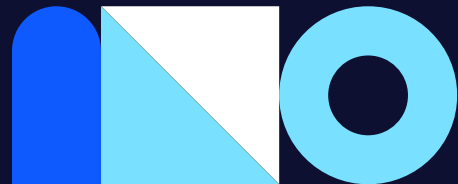
Always ask “compared to what?”

Follow post-quantum cryptography, try hybrid QML on a simulator, and read the assumptions behind any speedup claim.



References

- M. A. Nielsen & I. L. Chuang, Quantum Computation and Quantum Information, 10th Anniversary ed., Cambridge University Press, 2010 — §6.5 “Quantum search of an unstructured database” (pp. 265–268, Figs. 6.8–6.9); Ch. 7 (Fig. 7.7, p. 311).
- A. W. Harrow, A. Hassidim, S. Lloyd, “Quantum algorithm for linear systems of equations”, Physical Review Letters 103, 150502 (2009). — HHL
- V. Havlíček et al., “Supervised learning with quantum-enhanced feature spaces”, Nature 567, 209–212 (2019). — quantum kernels
- J. Biamonte et al., “Quantum machine learning”, Nature 549, 195–202 (2017).
- S. Aaronson, “Read the fine print”, Nature Physics 11, 291–293 (2015).
- E. Tang, “A quantum-inspired classical algorithm for recommendation systems”, STOC 2019 (arXiv:1807.04271, 2018).
- E. Farhi, J. Goldstone, S. Gutmann, “A Quantum Approximate Optimization Algorithm”, arXiv:1411.4028 (2014). — QAOA
- J. Preskill, “Quantum Computing in the NISQ era and beyond”, Quantum 2, 79 (2018).
- IBM Quantum hardware roadmap (ibm.com/quantum); Google Quantum AI, Willow announcement (Dec 2024).





Thank you

Questions?

Part 3 · Quantum × Big Data

